

SWYNEX_Task1_Data_Preparation

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[39] 0s

print("Final dataset shape:", clean_df.shape)

print("\nTotal missing values:", clean_df.isnull().sum().sum())

print("\nTotal duplicate rows:", clean_df.duplicated().sum())

print("\nFinal columns:")

print(clean_df.columns.tolist())

Final dataset shape: (891, 11)

Total missing values: 0

Total duplicate rows: 0

Final columns:

['PassengerId', 'Survived', 'Pclass', 'Name', 'Sex', 'Age', 'Sibsp', 'Parch', 'Ticket', 'Fare', 'Embarked']

[40] 0s

clean_df.to_csv("titanic_cleaned.csv", index=False)

print("Cleaned dataset saved successfully!")

Cleaned dataset saved successfully!

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Rows before removing duplicates: 891
Rows after removing duplicates: 891
Duplicates removed: 0

[29] ✓ 0s
age_median = clean_df["Age"].median()
print("Median Age:", age_median)
Median Age: 28.0

[30] ✓ 0s
clean_df["Age"] = clean_df["Age"].fillna(age_median)
print("Missing Age values:",
clean_df["Age"].isnull().sum())
Missing Age values: 0

[31] ✓ 0s
print(clean_df["Embarked"].mode()[0])
S

[32] ✓ 0s
embarked_mode = clean_df["Embarked"].mode()[0]
clean_df["Embarked"] = clean_df["Embarked"].fillna(embarked_mode)

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Missing values are identified before cleaning the dataset.

[24] ✓ Os

```
missing_values = df.isnull().sum()

print("Missing values in each column:")
print(missing_values)
```

...

Missing values in each column:

PassengerId	0
Survived	0
Pclass	0
Name	0
Sex	0
Age	177
SibSp	0
Parch	0
Ticket	0
Fare	0
Cabin	687
Embarked	2
dtype:	int64

3. Duplicate Record Analysis

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Pandas imported successfully!

[18] ✓ 0s
df = pd.read_csv("titanic.csv")
print("Dataset loaded successfully!")

Dataset loaded successfully!

[20] ✓ 0s
df.head()

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S

Next steps: [Generate code with df](#) [New interactive sheet](#)

[21] ✓ 0s
print("Number of rows and columns:", df.shape)

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